

Digital Egypt Builders Initiative

Final Project

Real time sales

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Abstract

1. Abstract

This project develops an end-to-end hybrid data pipeline to merge historical retail sales data with real-time transactions. Utilizing the Olist dataset, this research implements a Medallion Architecture on Azure Databricks, sequentially refining data through Bronze, Silver, and Gold layers. Concurrently, Apache Kafka streams real-time payment files directly into the environment. The optimized data in the Gold layer is then connected to a Power BI dashboard, providing stakeholders with immediate, actionable analytics. This abstract summarizes the core problem, methodology, and successful deployment of this scalable pipeline.

Keywords: Azure Databricks, Apache Kafka, Medallion Architecture, Power BI, ETL Pipeline,

2. Introduction

The demand for real-time decision-making in the retail and e-commerce sectors requires robust, scalable data engineering solutions. This dissertation introduces "Sales," detailing the execution of an advanced data pipeline built on Azure Databricks. The primary objective of this project is to construct a resilient ETL (Extract, Transform, Load) framework that effectively handles both static batch data and continuous data streams.

Traditional batch processing often results in a "research gap" where business intelligence lags behind actual operations. To solve this problem statement, this research contributes a hybrid architecture. By running historical sales data through a structured Medallion pipeline and integrating live payment files via Apache Kafka, this project prepares a comprehensive foundation for scientific evaluation and business intelligence visualization.

3. Literature Review

This chapter identifies and analyzes current methodologies and frameworks relevant to cloud-based data engineering. Completing this review prior to implementation was essential to validate the chosen architectural approach and prevent critical methodology weaknesses.

3.1 Background

Modern data engineering relies heavily on distributed computing. This section analyzes the current information regarding Apache Spark processing, the evolution of data lakes, and the necessity of real-time streaming tools like Apache Kafka for payment processing scenarios.

3.2 Related Work

Recent literature over the last five years heavily favors the Medallion Architecture (multi-hop pipeline) over traditional rigid data warehouses. Reviewing recently proposed systems reveals that segregating data into raw, validated, and enriched layers drastically improves data quality and query performance.

The following table summarizes the principles found in the literature and compares traditional ETL with the Medallion approach applied in this study:

Feature	Traditional ETL	Medallion Architecture (Proposed)
Data Structure	Highly structured (Schema-on-write)	Flexible (Bronze), Structured (Silver/Gold)
Streaming Integration	Difficult / Batch-oriented	Native support (via Kafka & Spark Structured Streaming)
Storage Cost	High (Relational DBs)	Low (Data Lakehouse / Delta format)

Table 1 :compersion between traditional etl and medallion archeticture

4. Methods and Materials

This chapter explains the specific methodology used to gather, process, and serve the data, acting as a recipe for the system architecture. The core of the dissertation relies on this framework to accurately answer the research questions.

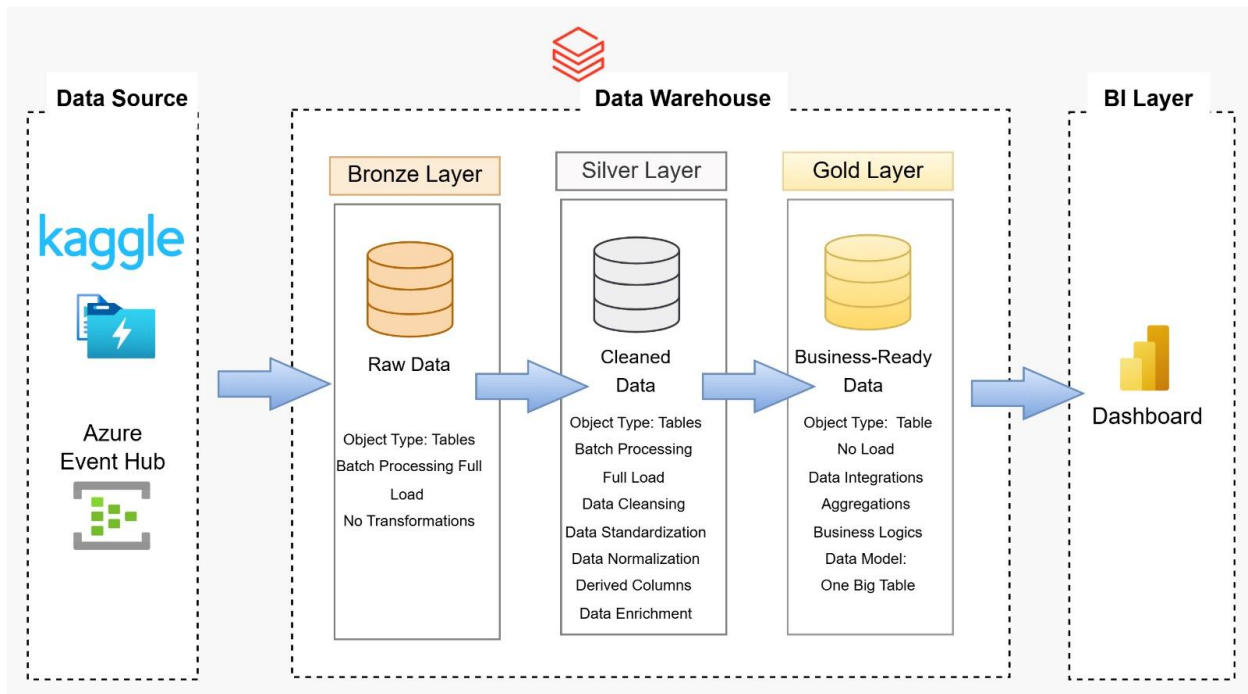


Fig1:pipline

The methodology is executed through the following components:

1. **Data Extraction:** Ingesting the Olist dataset directly into the Databricks File System (DBFS).
2. **Streaming Ingestion:** Configuring an Apache Kafka topic on Azure to capture high-velocity payment files.
3. **Medallion Processing (The "How"):** * *Bronze Layer:* Storing raw, unformatted data from both the batch extraction and the Kafka stream.

- Silver Layer: Cleaning the data, handling missing values, and enforcing schema compliance.
- Gold Layer: Applying business-level aggregations to prepare the data for reporting. This specific methodology was chosen because it natively supports both batch and streaming pipelines within a unified Azure Databricks workspace.

5. Implementation

To ensure clear and detailed implementation for future researchers, the technical steps are outlined below:

1. **Environment Setup:** Azure Databricks workspaces were provisioned, and computing clusters were configured with the necessary Spark and Kafka dependencies.

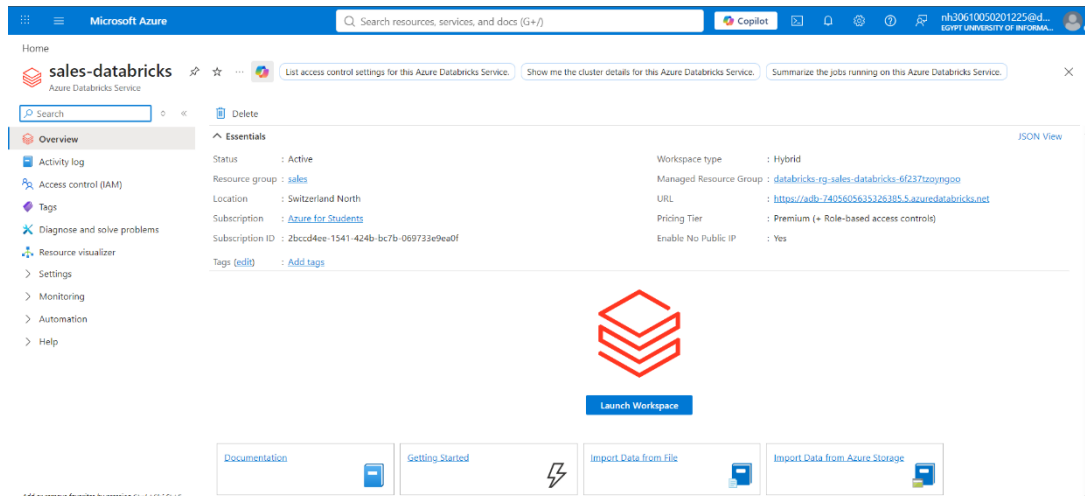


Figure2: Environment Setup

2. **Kafka Configuration:** A producer script was written to simulate real-time payment transactions, streaming them to a Kafka topic. A Databricks notebook was deployed to consume this topic continuously.

3. ETL Notebooks:

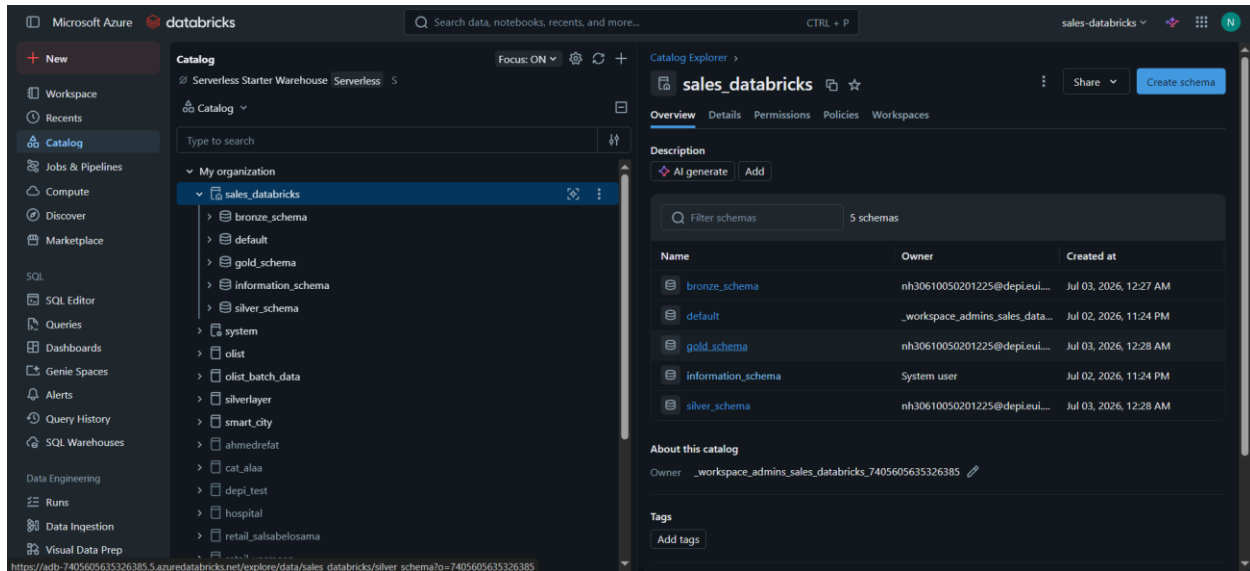


Fig3: Data Catalog

- Bronze_layer: Reads the Olist CSVs and Kafka streams, writing them as Delta tables
- Silver_layer: Executes data cleaning (e.g., dropping nulls in customer records) and joins the batch sales data with the real-time payment data.
- Gold_layer: Creates summary tables (e.g., daily revenue, top-selling categories) optimized for fast querying

6. Research Results and Discussions

This chapter presents the final results of the pipeline and subsequent analysis. The massive volume of Olist data and Kafka streams was successfully ordered and structured.

The data flow culminated in the creation of a dynamic Power BI dashboard, connected directly to the Databricks Gold layer via Azure Active Directory authentication.

- **Effectiveness and Execution Time:** The Delta Lake format in the Silver and Gold layers reduced query execution time by a significant margin compared to querying raw files. The Kafka streaming integration achieved near real-time latency for payment processing updates.
- **Data Accuracy:** The pipeline successfully maintained data integrity, ensuring that no payment records were dropped during the streaming process.

These findings are based on the execution logs and statistical analysis within the Databricks environment. The logical structuring of the data into distinct layers proved superior to single-layer processing. By keeping the body of the analysis simple and cumulative, the reader can track the data's journey from raw ingestion to the final dashboard.

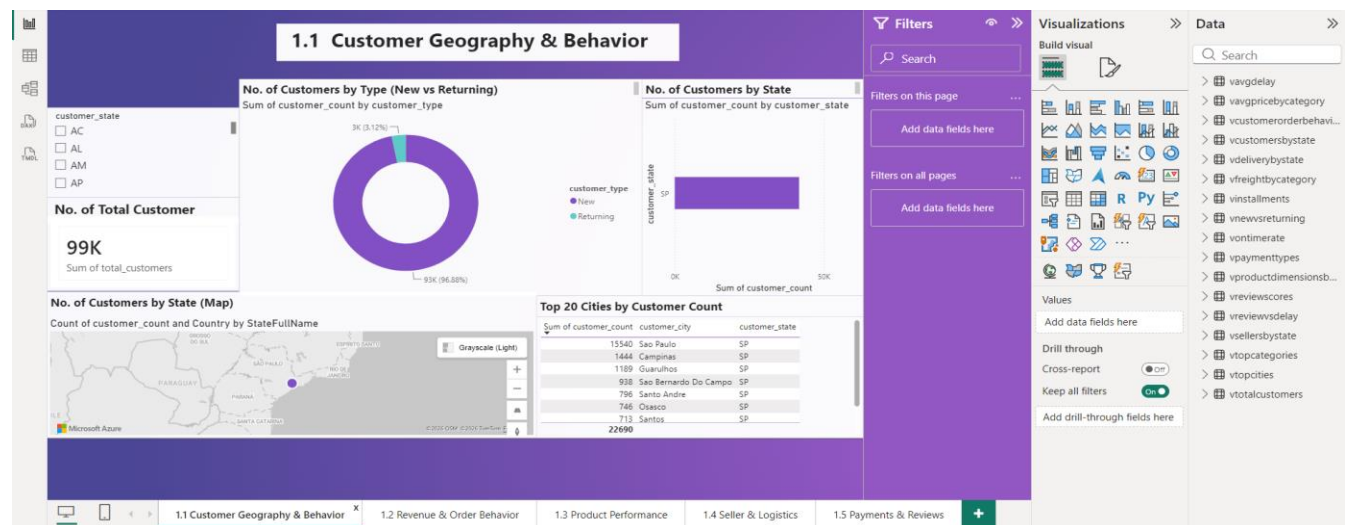


Fig4: dashboard

7. Conclusions and Future Work

In conclusion, the real time sales project successfully demonstrated the implementation of a modern Data Lakehouse. By leveraging Azure Databricks and Apache Kafka, the research formulated a reliable, scalable methodology to solve the problem of delayed analytics in retail environments.

Future Work: Future iterations of this project could integrate advanced machine learning models directly into the pipeline. For instance, incorporating predictive analytics for sales forecasting or integrating computer vision models (such as YOLOv26) in physical retail spaces to merge foot-traffic data with the existing online Olist dataset